Learning Causally Predictable Outcomes from Psychiatric Longitudinal Data

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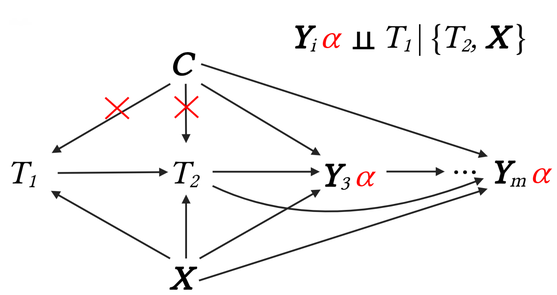
Full paper:

Abstract

- **Problem:** In psychiatric longitudinal data, fixed outcomes with covariate adjustment often remain biased due to latent confounding.
- **Idea:** *Learn the outcome definition.* Estimate symptom weights that render treatment effects identifiable and durable over time.
- **Method:** *DEBIAS* learns non-negative item weights that (i) maximize partial correlation with current treatment and (ii) penalize latent confounding by leveraging time-limited direct effects.
- **Empirical test:** An empirical check of projected unconfoundedness via partial-correlation p-values.
- **Results:** In TADS (depression) and CATIE (schizophrenia), DEBIAS improves out-of-sample partial correlation and raises confounding-test p-values relative to alternative algorithms.
- **Open Source:** `https://github.com/ericstrobl/DEBIAS`

Introduction

- **Challenge:** Latent confounders violate ignorability for total scores and remission rates.
- **Key prior:** Many historical treatments have *time-limited direct effects*; long-run effects are mediated by current treatment.
- **Approach:** Project outcomes onto a non-negative combination that is conditionally independent of past treatments given current treatment and covariates.
- **Benefit:** Restores unconfoundedness for the learned score, enabling valid causal estimates.
- **Interpretability:** Non-negativity yields clinically meaningful severity composites (no cancellation).



Prioritizing Causal Differentiation via Correlation

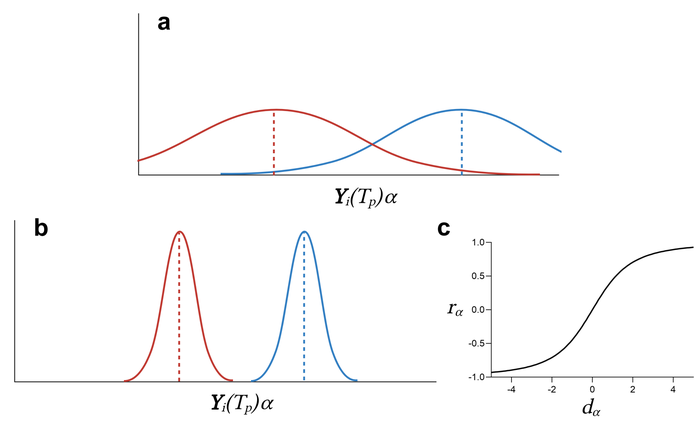


Figure 1. **CATE vs. Cohen's d vs. correlation.** (a) **Maximizing CATE** can yield large within-group variance, so treated/control distributions still overlap. (b) **Maximizing Cohen's d** standardizes the mean difference by the pooled SD, improving separability—but d only handles binary T. (c) **Why correlation?** For binary T, it moves monotonically with d. Unlike d, correlation naturally handles ordinal and continuous treatments, so DEBIAS maximizes **correlation**.

Durable Effects with Backdoor-Invariant Aggregated Symptoms (DEBIAS)

- **Goal:** Learn severity scores $\mathbf{Y}_i\alpha$ ($\alpha \geq 0, \|\alpha\|_1 = 1$) that are *durably* linked to current treatment T_p and *empirically unconfounded*.
- **Objective (first score):**

$$\underset{\alpha\geq 0,\ \|\alpha\|_1=1}{\arg\max} \sum_{i=p+1}^{m} \Big[\mathrm{cor}(\mathbf{Y}_i\alpha, T_p \mid T_1, \mathbf{X}) - \frac{\lambda}{p-1} \sum_{j=1}^{p-1} \mathrm{cor}^2(\mathbf{Y}_i\alpha, T_j \mid T_p, \mathbf{X}) \Big]$$

- **Multiple scores:** Add a correlation-weighted cosine similarity penalty to diversify new α's across symptom axes.
- **Inference check:** Require squared partial correlation p-values (with past T_j) to exceed a threshold (default 0.05).
- **Optimization:** Projected gradient ascent with Armijo backtracking; cross-validate λ.

Results

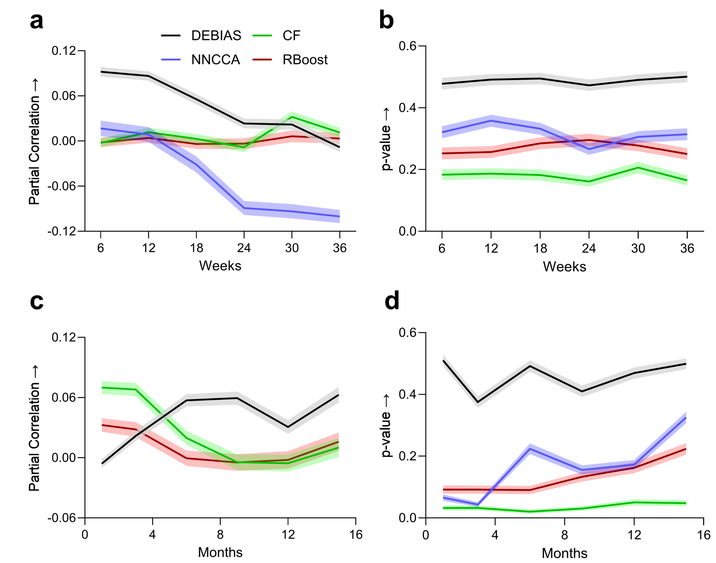


Figure 2. **Empirics in depression (TADS) and schizophrenia (CATIE).** (a) **TADS:** DEBIAS attains higher partial correlation with T_p across future weeks when treatment effects diverge. (b) **TADS:** DEBIAS yields larger confounding-test p-values, indicating stronger control of latent confounding. (c) **CATIE:** DEBIAS achieves higher partial correlation as antipsychotic effects diverge over months. (d) **CATIE:** DEBIAS shows larger confounding-test p-values, reflecting improved confounding control. Bootstrapped 95% CIs are tight; DEBIAS outperforms NNCCA+IPTW, R-Learner, and Causal Forests.

Conclusions

- **Outcome learning for causality:** Choosing *which* outcome to analyze can unlock identification even when fixed totals and remission rates fail.
- **Empirical unconfoundedness:** DEBIAS provides a falsifiable check using partial-correlation tests.
- **Clinical utility:** Non-negative, durable composites map to symptom domains and are easy to communicate.
- **Scope:** Linear, fast, and effective; extends to continuous/ordinal treatments.
- **Limits & next steps:** Partial decorrelation $\not\equiv$ full independence; explore nonlinear dependence, broader applications to other areas in medicine.